



GCE A LEVEL

1420U50-1D



S26-1420U50-1D

To be opened on WEDNESDAY, 18 MARCH 2026 by TEACHERS

PHYSICS – A2 unit 5
Practical Examination

Experimental Task
TESTS 1 and 2

SETTING UP INSTRUCTIONS

CONFIDENTIAL

This document should be stored securely by the exams officer when not in use by the teacher/technician.

This document must not be stored within the science department when not in use by the teacher/technician.

Its contents should not be divulged except to those concerned with the preparation of the assessment.

1420U501D
01

Answer all questions.

You are going to investigate how the **amplitude** of vertical vibration, A , of a mass hanging from two springs in series varies with time, t .

- (a) The relationship between the amplitude of vibration, A , and the time, t , can be expressed as:

$$A = A_0 e^{-\lambda t}$$

where A_0 = initial amplitude and λ = unknown constant.

Your task is to design and carry out an experiment to determine λ .

You are provided with the following equipment:

500g mass including hanger
2 linked springs
split cork (optional)
2 clamp and stand
G-clamp
metre ruler
stopwatch ± 1 s

Write a plan of how you will obtain sufficient readings to find a value for λ . You will need to include a labelled diagram, risk assessment and explanation of how you will use your results to draw an appropriate graph. [8]

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Risk assessment

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- (b) Using the apparatus, take sufficient measurements to complete this task. Record your results below, **all values of amplitude should be in centimetres**. State the resolutions of all instruments used. [4]

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The remaining parts of the examination paper are not required for the purpose of checking the setting up of the task.

TEST 2

All of the above applies except **mass of 100g** is used.

AWARDING OF TEACHER ASSESSED MARKS

Please ensure that these marks are recorded on the front of each candidate's paper.

- (a) (i) **The mark is to be awarded if the candidate has identified the graph to plot as:**
 $\ln A$ against t . Accept $\log A$ against t .

If the candidate is incorrect **do not award the mark** but give the candidate the information sheet (on the next page) to tell them the correct graph to plot.

However, if a candidate identifies the graph to plot as t against $\ln A$ and the correct method to find λ and A_0 is given then this is acceptable too. The information sheet **should not** be given in this case.

- (ii) **This mark is to be awarded if the candidate states that there is no significant risk. Accept risk of clamp stand falling over.**

INFORMATION SHEET

**Sufficient copies of this information sheet should be made in advance.
The information sheet is the same for Tests 1 and 2.**

INFORMATION SHEET

The graph that should be plotted is $\ln A$ on the y -axis and t on the x -axis.

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